

Towards Design of Human-Centered AI Systems

Motahhare Eslami

A part of this presentation has
been adopted from Min Kyung Lee

Research shows more than 80% of AI projects fail, wasting billions of dollars in capital and resources: Report

News

By [Jowi Morales](#) published August 28, 2024

The AI industry is fraught with failure.



Comments (20)



Research shows more than 80% of AI projects fail, wasting billions of dollars in capital and resources: Report

News

By [Jowi Morales](#) published August 28, 2024

According to the research, the biggest reason for the failure of AI projects is the misalignment of goals between key stakeholders. The leadership often has a view of what AI can and should achieve that is not grounded in reality; instead, it's driven by humanity's pre-conceived notion of what AI is, most often fuelled by Hollywood. This lack of understanding between business leaders and the people on the ground means that projects often do not have the resources and time needed to accomplish their goals.

Remember Module 1?

Module I: Introduction & Guidelines

Session 1: Introduction to Human-AI Interaction

Session 2: Practical Machine Learning

Session 3: Perspectives on Human-AI Interaction

Session 3 Reading Reflection
Sep 3 | 1 pts

Session 4: Guidelines of HAI

Session 4 Reading Reflection
Sep 8 | 1 pts

Session 5: AI in Social Decision-making Contexts

Session 5 Reading Reflection
Sep 10 | 1 pts

Session 6: Designing for AI Failures

Session 7: Stakeholder-driven and Value-Sensitive AI Design

Session 7 Reading Reflection
Sep 17 | 1 pts

Assignment 1
Sep 19 | 10 pts

User-centered design process

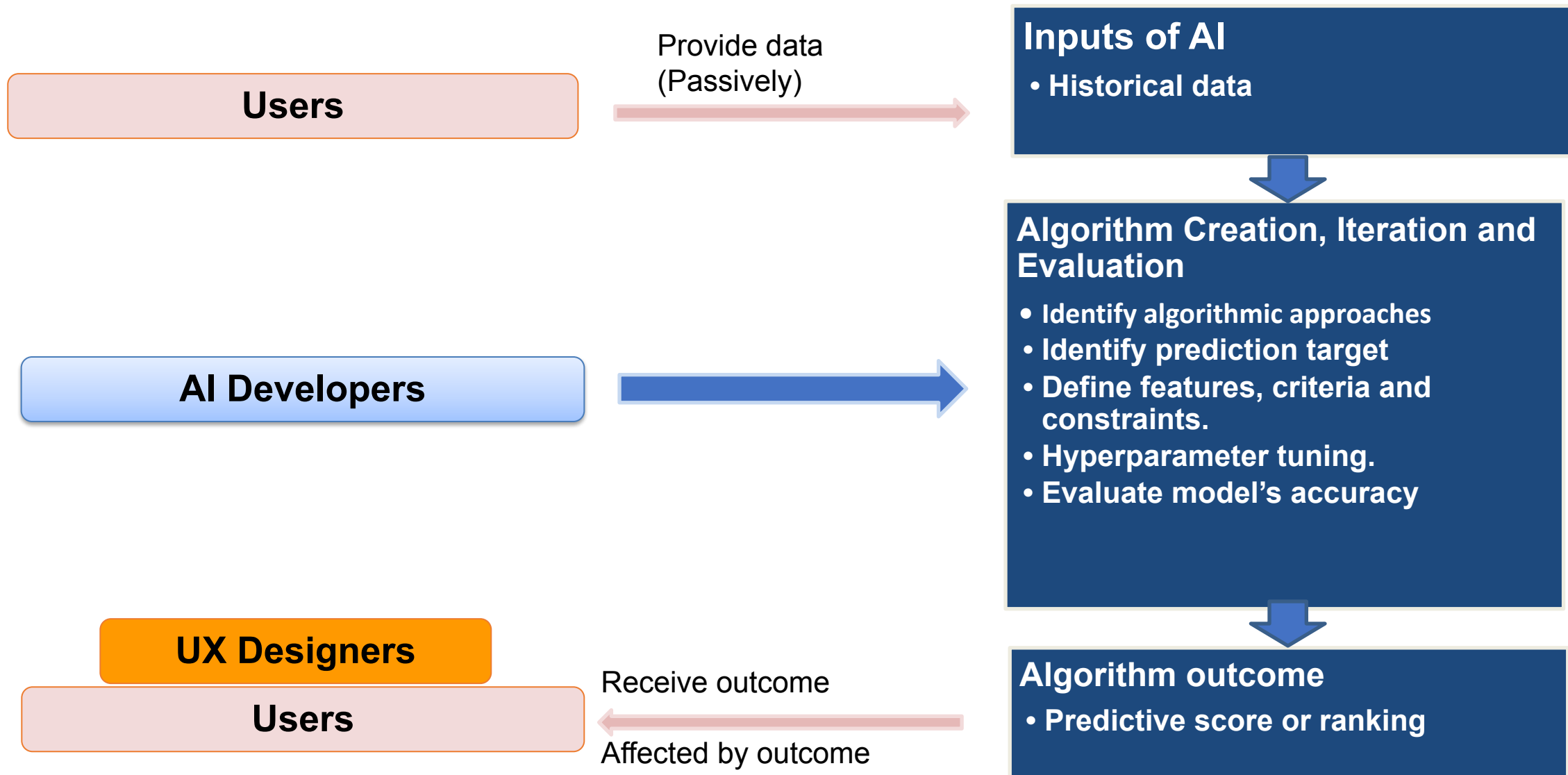
- **Consider users throughout the process of design and development in a principled and comprehensive manner.**
- **Incorporates users' considerations and constraints** into early stages of design.
- **User-centered evaluation**



“If I could wave a magic wand, I would use it to make the HCI community **move beyond user-centered design** to a notion of **stakeholder-centered design**.”

- Jodi Forlizzi (2018)

Traditional development approach



Stakeholder-centric approach

UX Designers

Researchers

Stakeholders

AI Developers

Researchers

Stakeholder control in data sharing
and protect stakeholder privacy

Use stakeholders' inputs to guide the specific
algorithmic choices and decision-making

Consider the social and organizational
contexts

Iterative and open refinement with h stakeholders

Evaluate accuracy, user experience, acceptability,
& impacts

Explain the outcomes (Interpretability)

Continuous evaluation

Inputs of AI

- Historical data

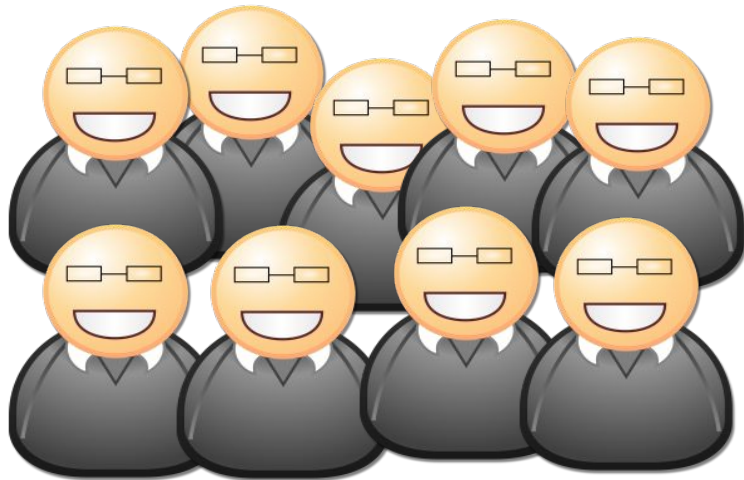
Algorithm Creation, Iteration and Evaluation

- Identify algorithmic approaches
- Identify prediction target
- Define features, criteria and constraints.
- Hyperparameter tuning.
- Evaluate model's accuracy

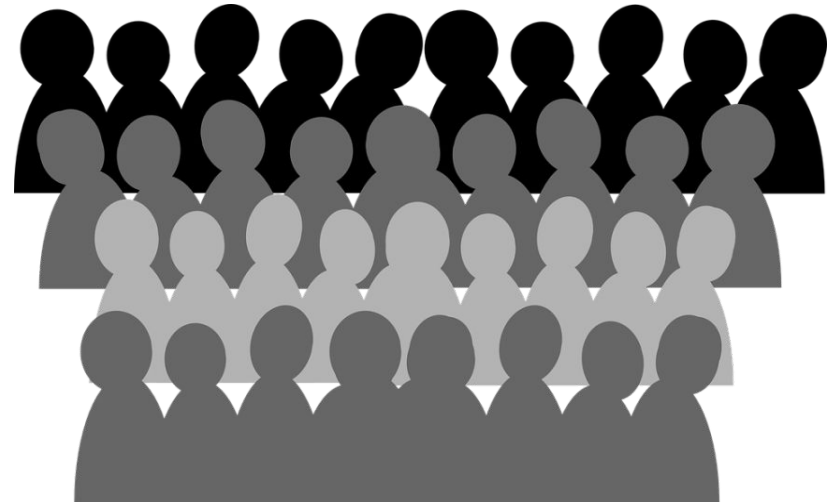
Algorithm outcome

- Predictive score or ranking

Lack of *Meaningful* Stakeholders' Engagement, *Early* and *Throughout* the process of AI Development



VS



Human-Centered AI Design

Empowering **users and impacted stakeholders** of AI systems, especially **those whose voices are often unheard**, to be engaged and have a say in the design, development, and evaluation of accountable systems

Human-Centered AI Design



WeBuildAI: Collective participatory framework

Enable collective participation of stakeholders in defining algorithms

Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

WeBuildAI: Collective participatory framework

Build a computational model representing each individual stakeholder and to have that model vote on its individual's behalf

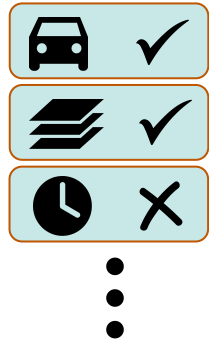
Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

WeBuildAI: Collective participatory framework

Build a computational model representing each individual stakeholder and to have that model vote on its individual's behalf

Step 1

Feature selection

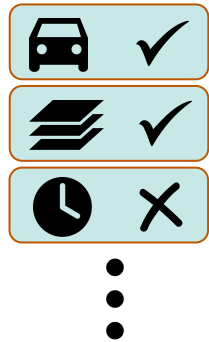


Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

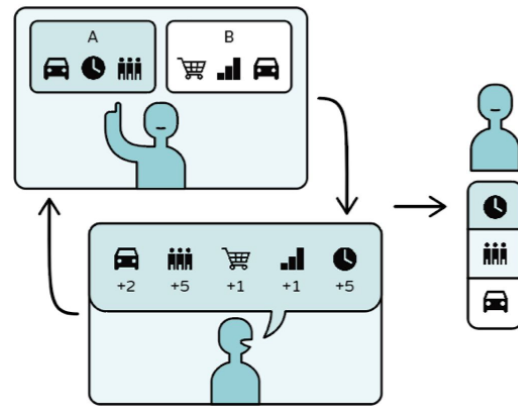
WeBuildAI: Collective participatory framework

Build a computational model representing each individual stakeholder and to have that model vote on its individual's behalf

Step 1
Feature selection



Step 2
Individual modeling

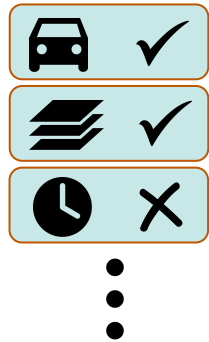


Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

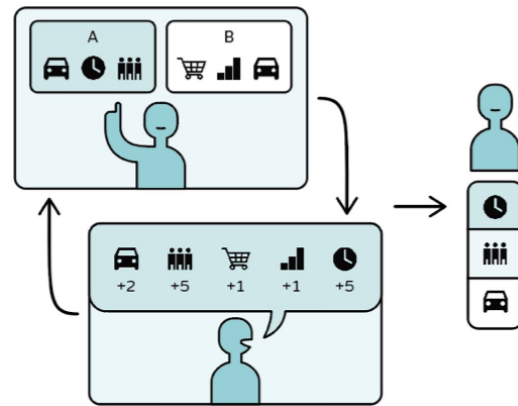
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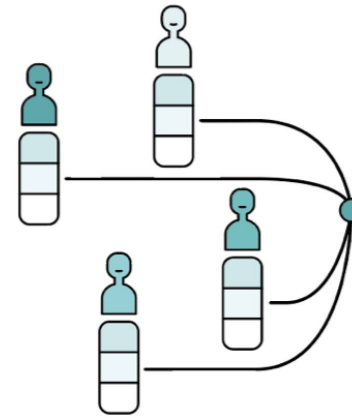
Step 1
Feature selection



Step 2
Individual modeling



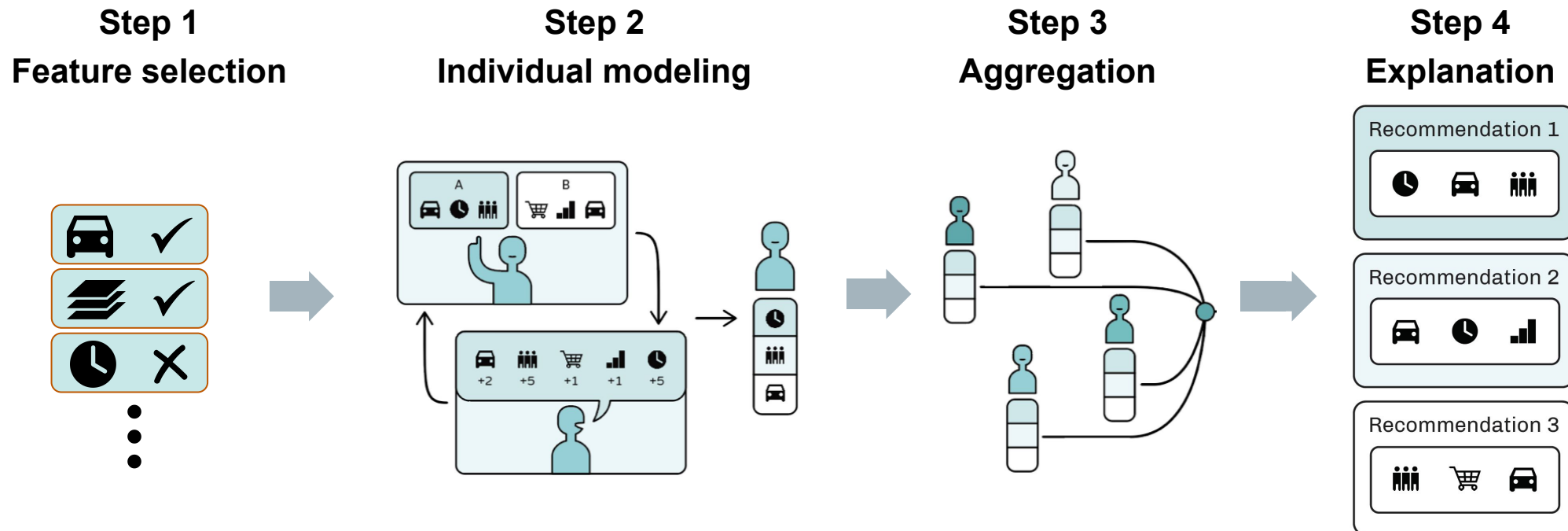
Step 3
Aggregation



Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

WeBuildAI: Collective participatory framework

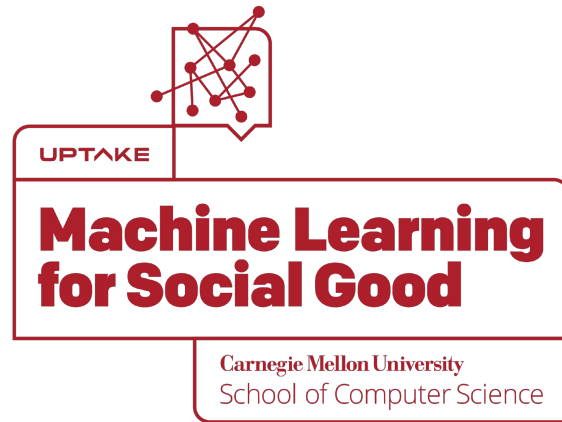
Build a computational model representing each individual stakeholder and to have that model vote on its individual's behalf



Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Application context

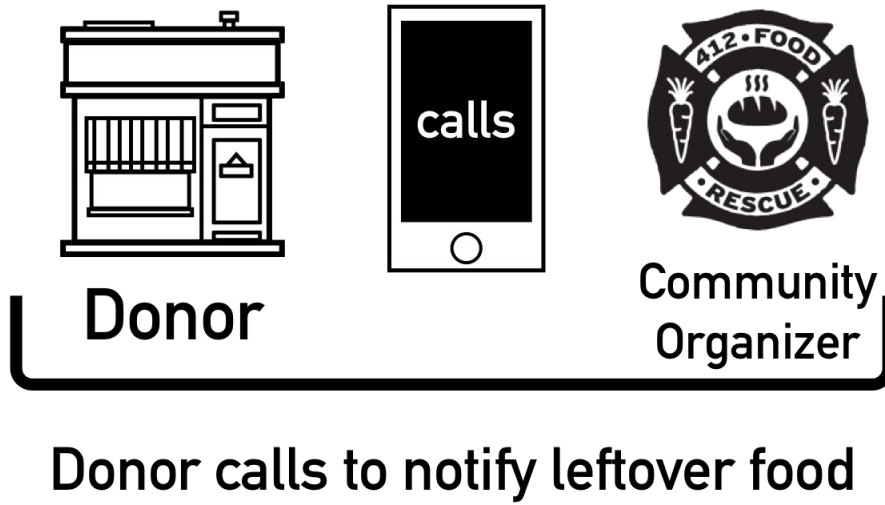
An on-demand donation transportation service provided by 412 Food Rescue



Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Application context

An on-demand donation transportation service provided by 412 Food Rescue

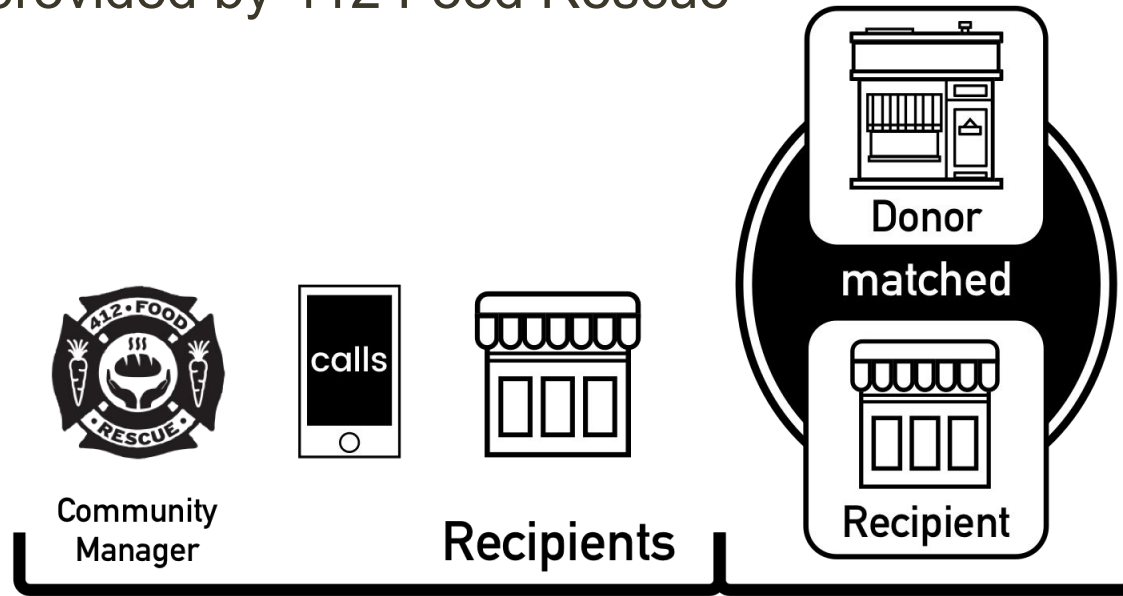


TARGET



Application context

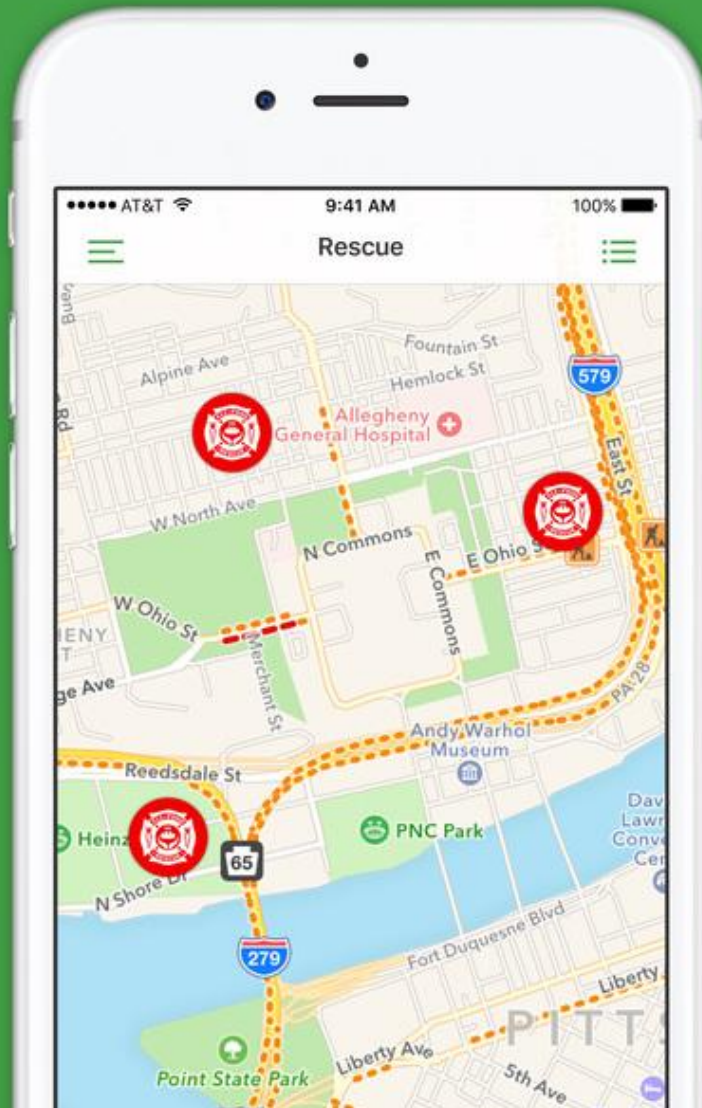
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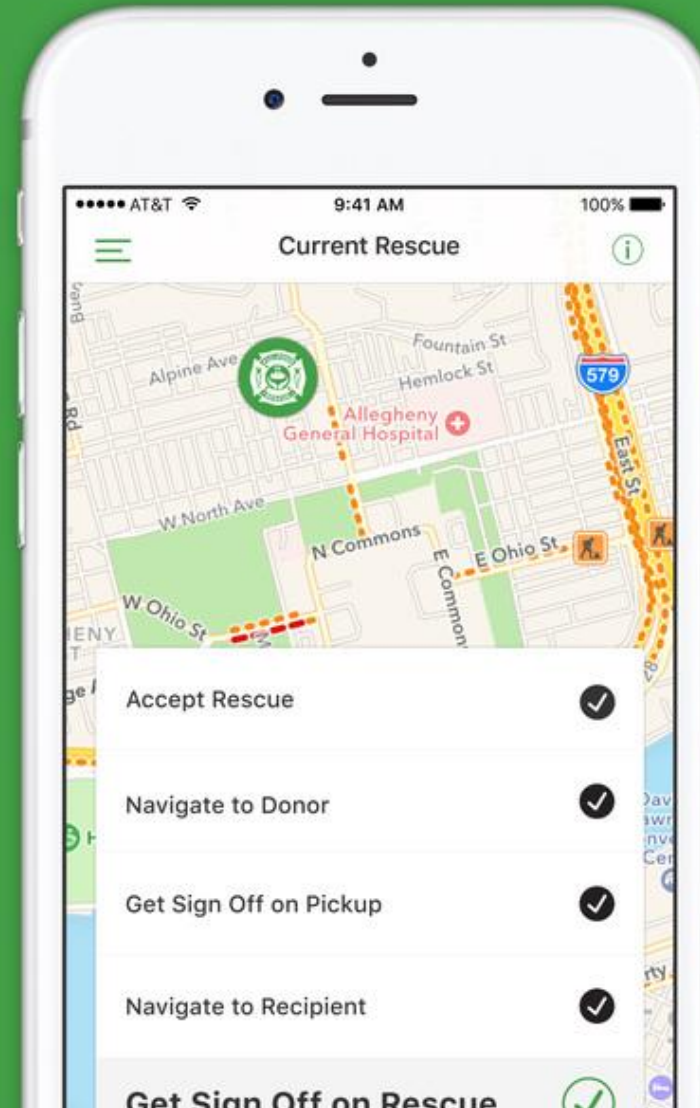
412 Food Rescue Coordinates Allocation

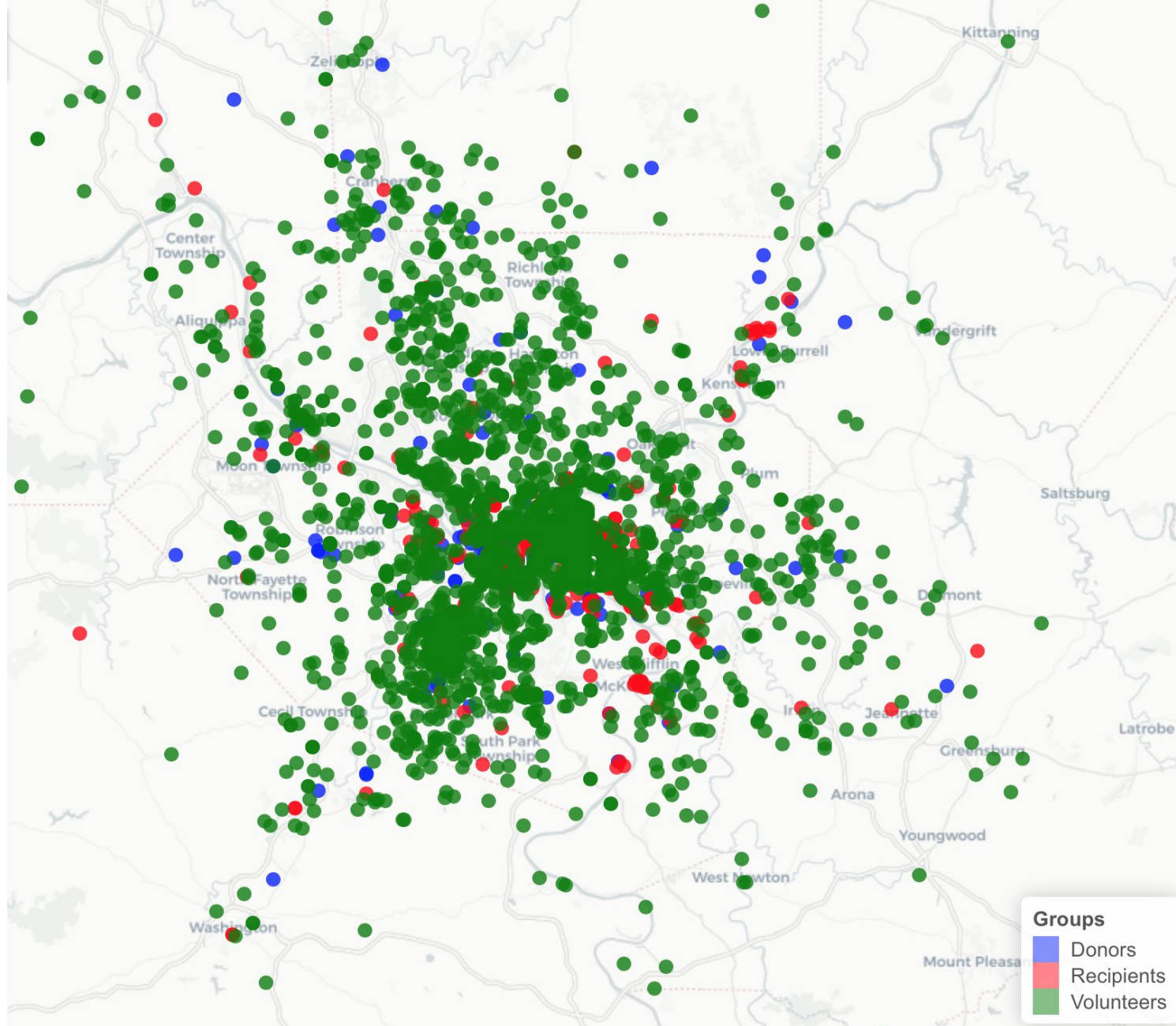


See rescues when you're
available



Easy step-by-step information
on pick up and drop off

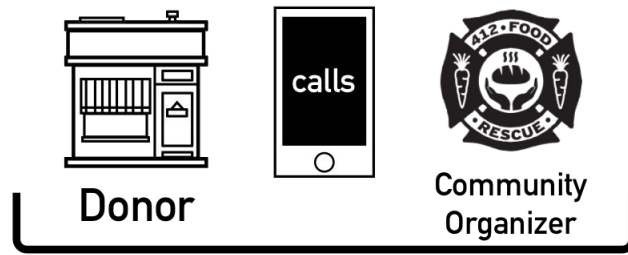




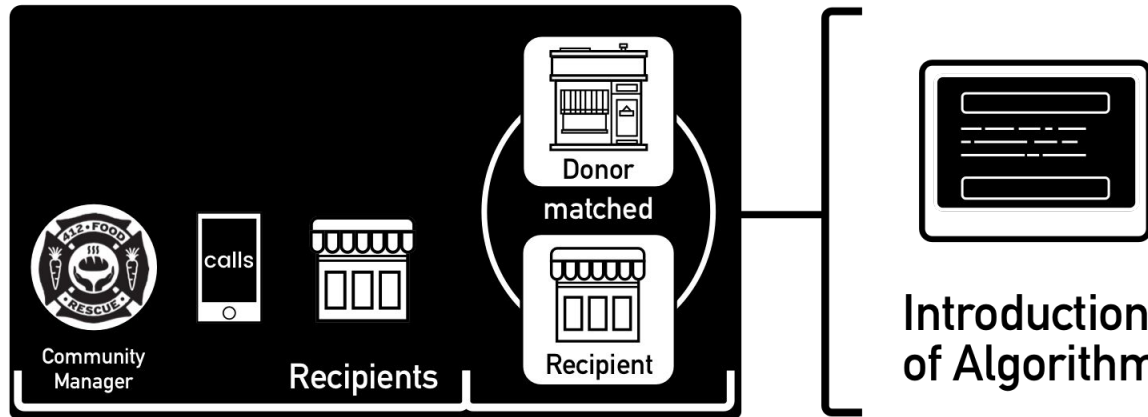
3 million pounds of rescued food

Expanding their model into organizations in other cities





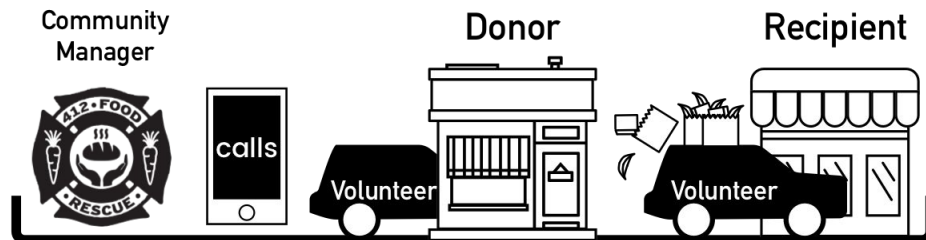
Donor calls to notify leftover food



412 Food Rescue Coordinates Allocation

Goals:

- Offload workload
- Improve equity



412 Food Rescue
Recruits Volunteers

Volunteers transport donation
from donor to recipient

Balancing equity & efficiency (1/2)

Collective participation to determine the tradeoff between equity and efficiency

It requires a normative approach

A human needs to define how to balance these competing factors

There is no fundamental “right” balance and cannot be determined purely through optimization

Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Balancing equity & efficiency (2/2)

Different impacts of algorithm design choice on different stakeholders

Efficiency: Shorter rescue

- Volunteer



Recipient



412 FR



Donor



Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Balancing equity & efficiency (2/2)

Different impacts of algorithm design choice on different stakeholders

Efficiency: Shorter rescue

- Volunteer	Recipient	412 FR	Donor
			

Equity: Recipient in greater needs

- Volunteer	Recipient	412 FR	Donor
			

Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Balancing equity & efficiency (2/2)

Different impacts of algorithm design choice on different stakeholders

Efficiency: Shorter rescue

- Volunteer	Recipient	412 FR	Donor
			

Equity: Recipient in greater needs

- Volunteer	Recipient	412 FR	Donor
			

Finding a collective solution to this tradeoff is critical to the success of the service

Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Method

Used the framework to build the allocation algorithm with stakeholders

Volunteer

V1: White male 60s
V2: White female 30s
V3: White female 60s
V4a: White female 70s
V4b: White male 70s
V5: White female, 60s
V6: White female, 20s

Recipient

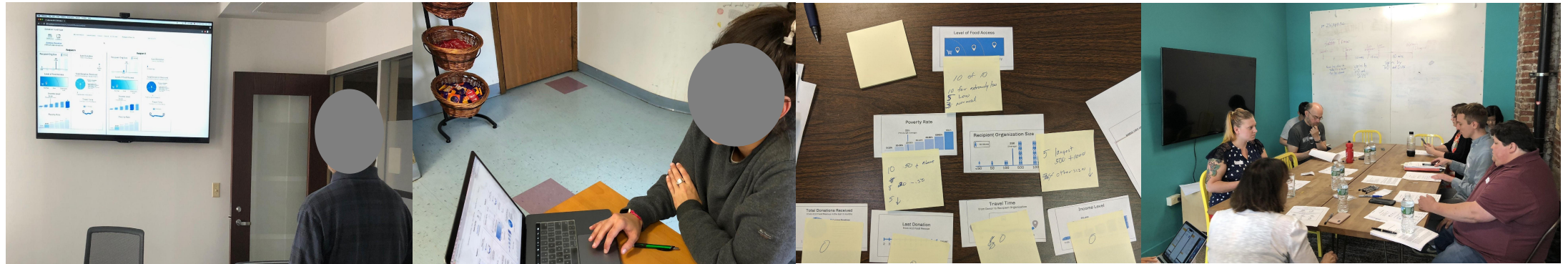
R1: Human Services Program manager
R2: Shelter & Food Pantry Center director
R3: Food Pantry staff
R4: Animal Shelter staff
R5: Food Pantry staff
R6: After school program employee
R7: Home-delivered meals manager
R8: Food Pantry director

412 FR

F1: CEO
F2: Staff1
F3: Staff2

Donor

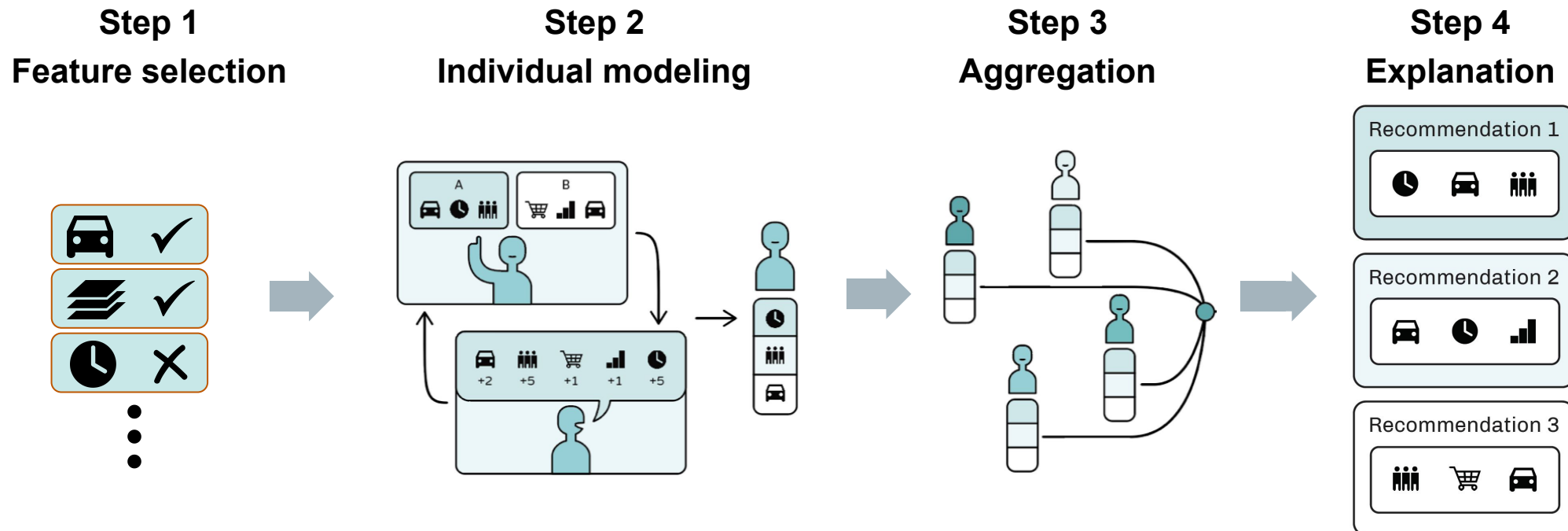
D1: School A
D2: School B
D3: Produce company
D4: Grocery
D5: Catering service
D6: School C



Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

WeBuildAI: Collective participatory framework

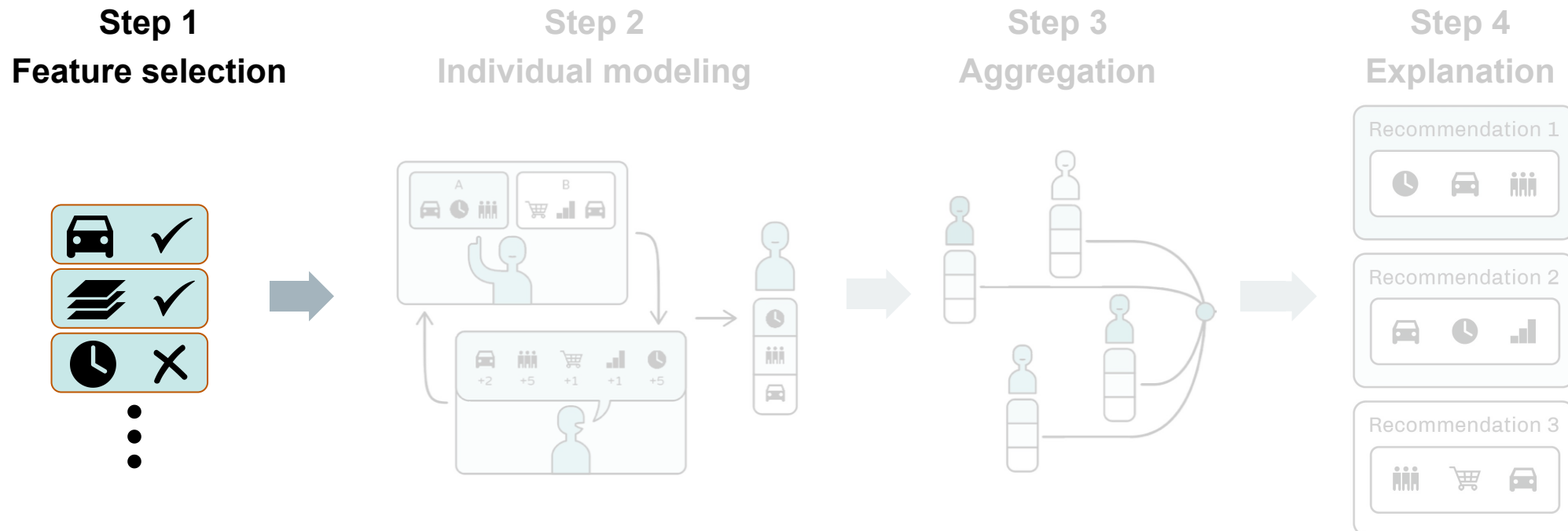
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Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

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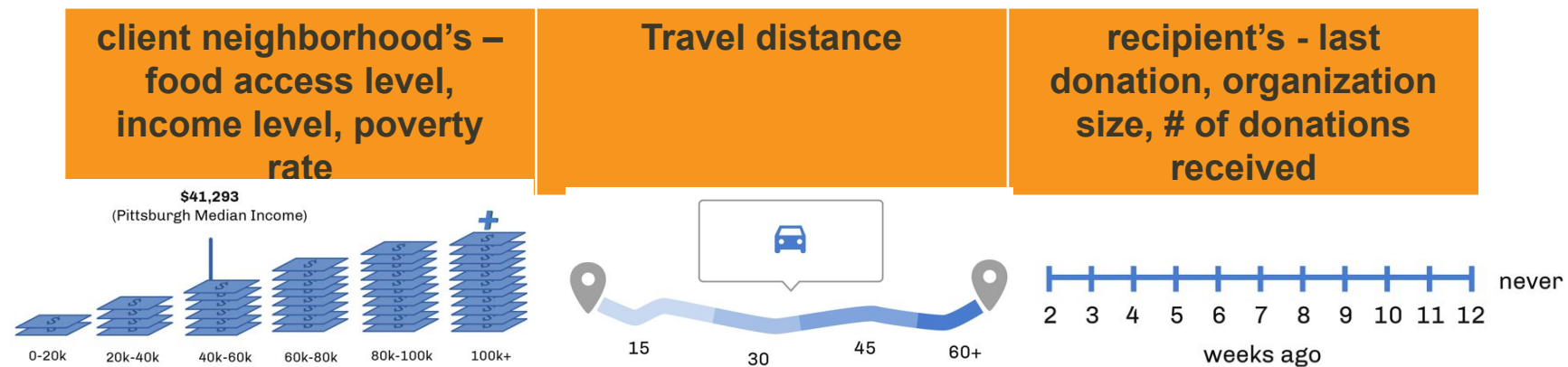


Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Step1: Feature selection

Conducted formative interviews with the stakeholders

What information should be used for allocation?

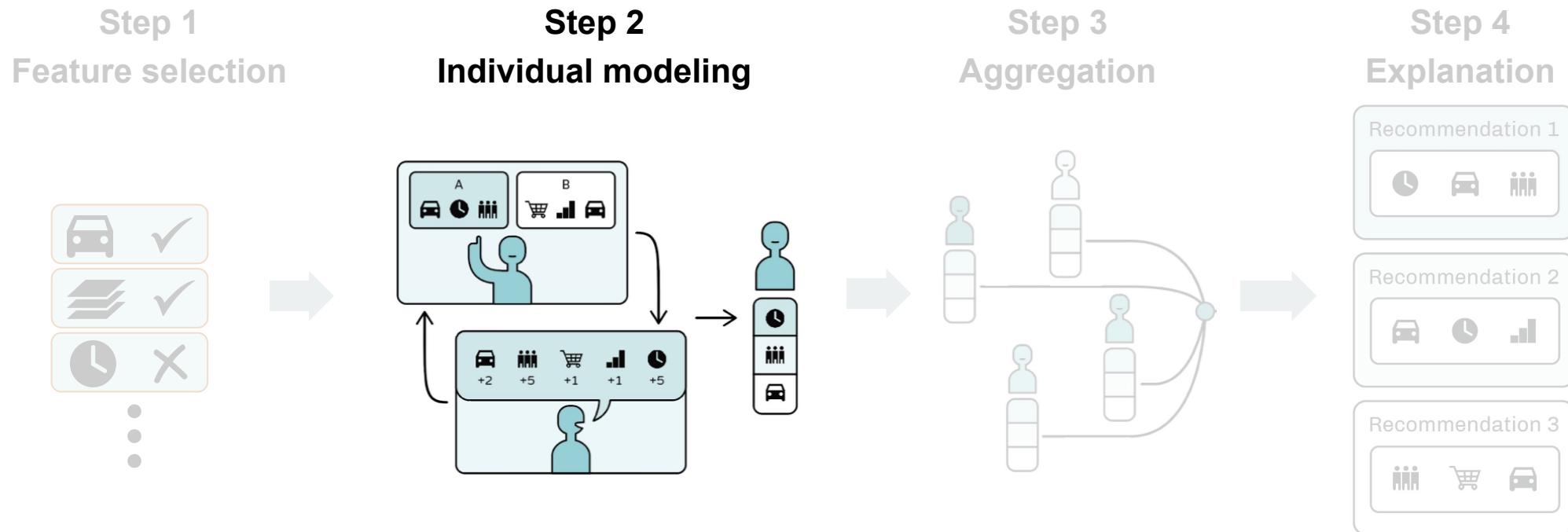


Selected the ones that most people mentioned and that had reliable data sources

Lee, Kim & Lizarondo. (2017). A human-centered approach to algorithmic services. CHI2017

WeBuildAI: Collective participatory framework

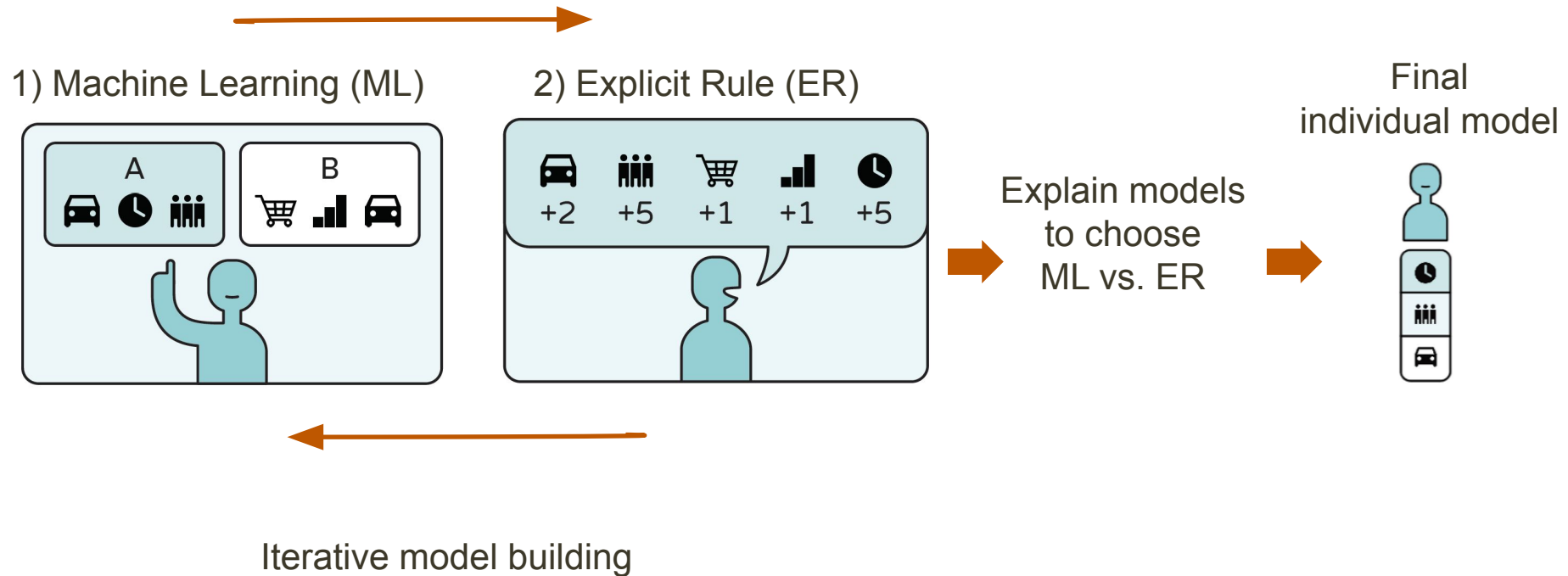
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Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Step2: Individual model building

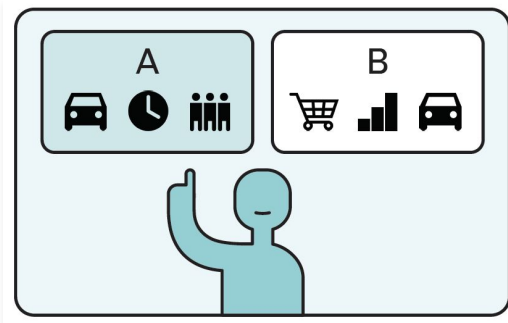
Used two methods iteratively to model individual's belief on how the algorithm should determine recipients



Step2: Individual model building

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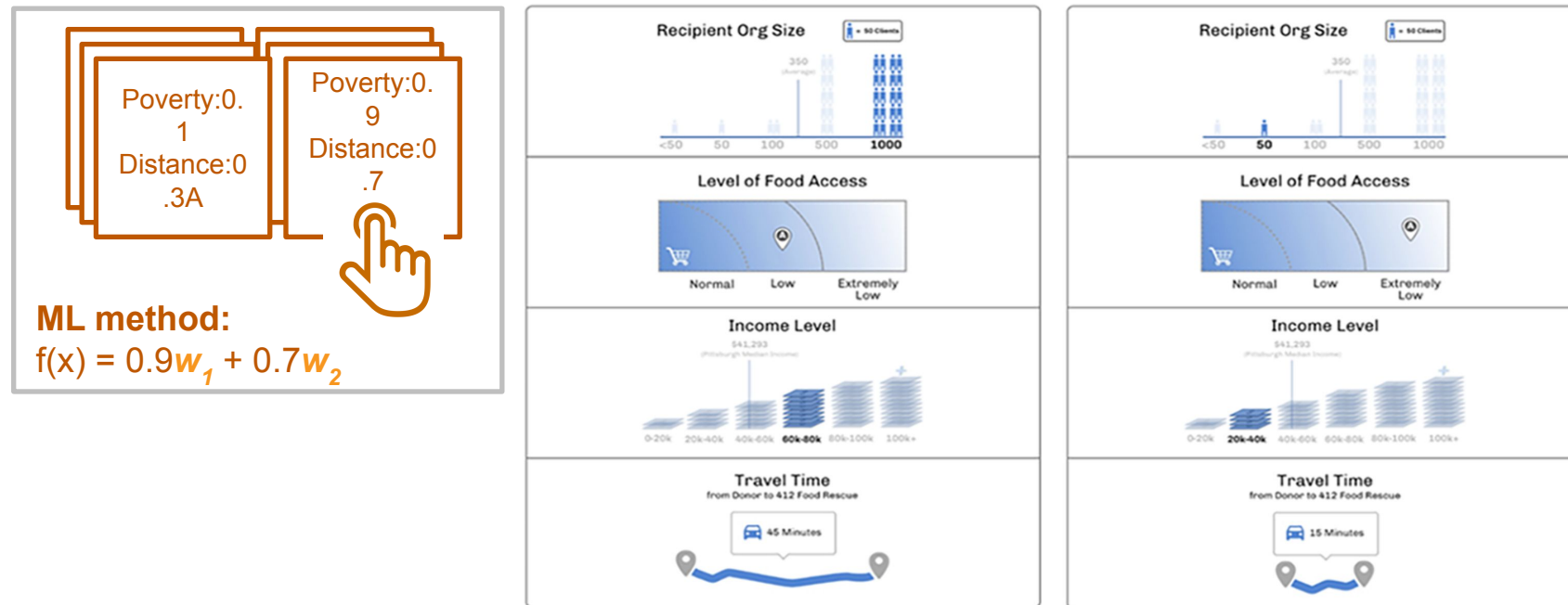
1) Machine Learning (ML)



Iterative model building

Individual belief elicitation

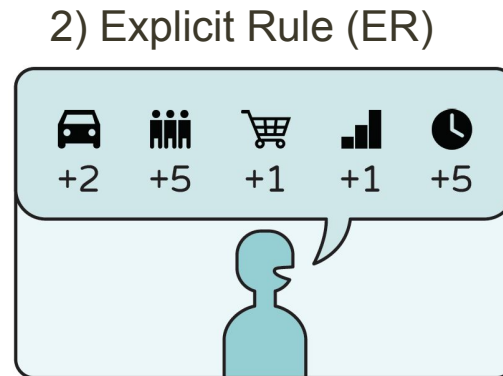
Machine learning method



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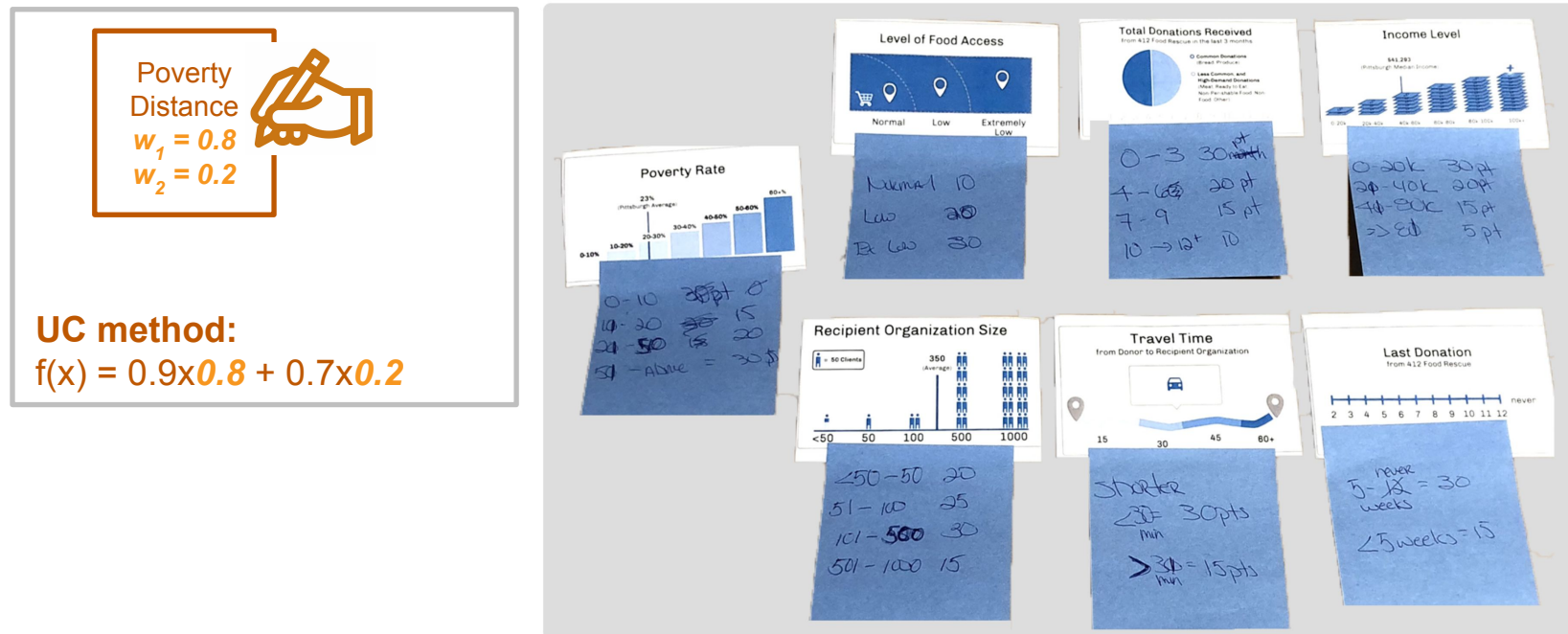
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Iterative model building

Individual belief elicitation

Explicit rule specification method



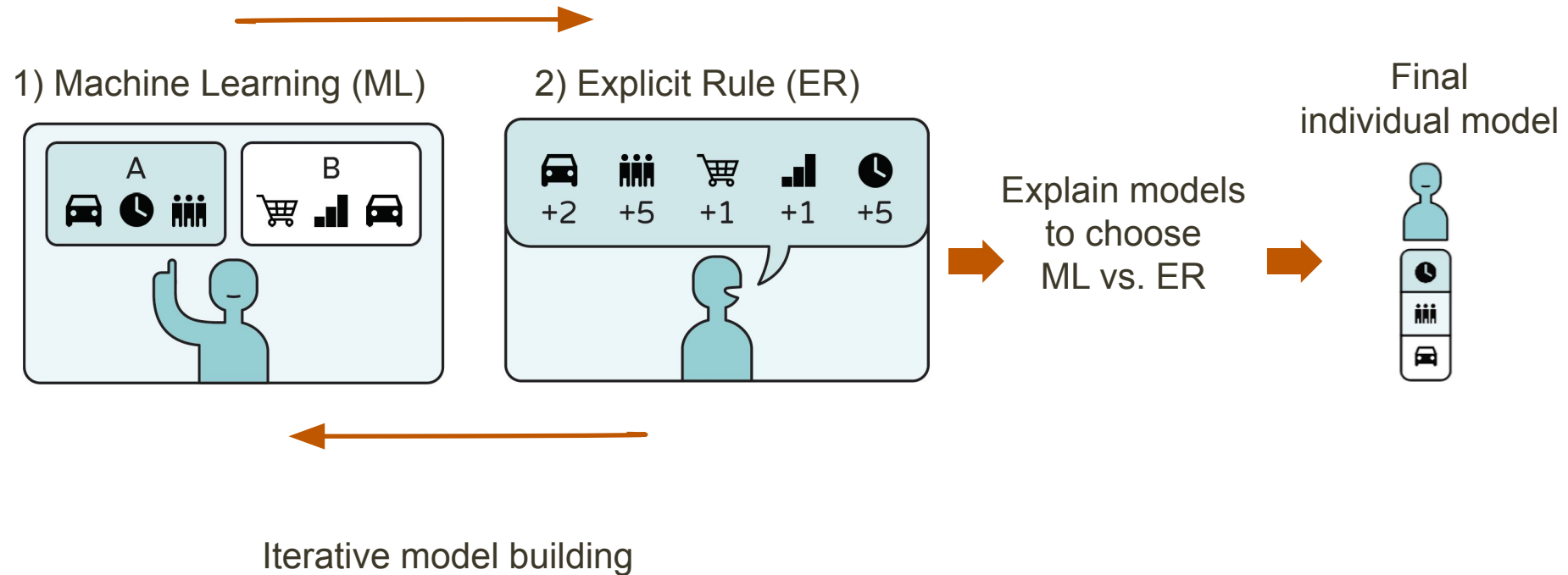
Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Machine learning vs explicit rule models



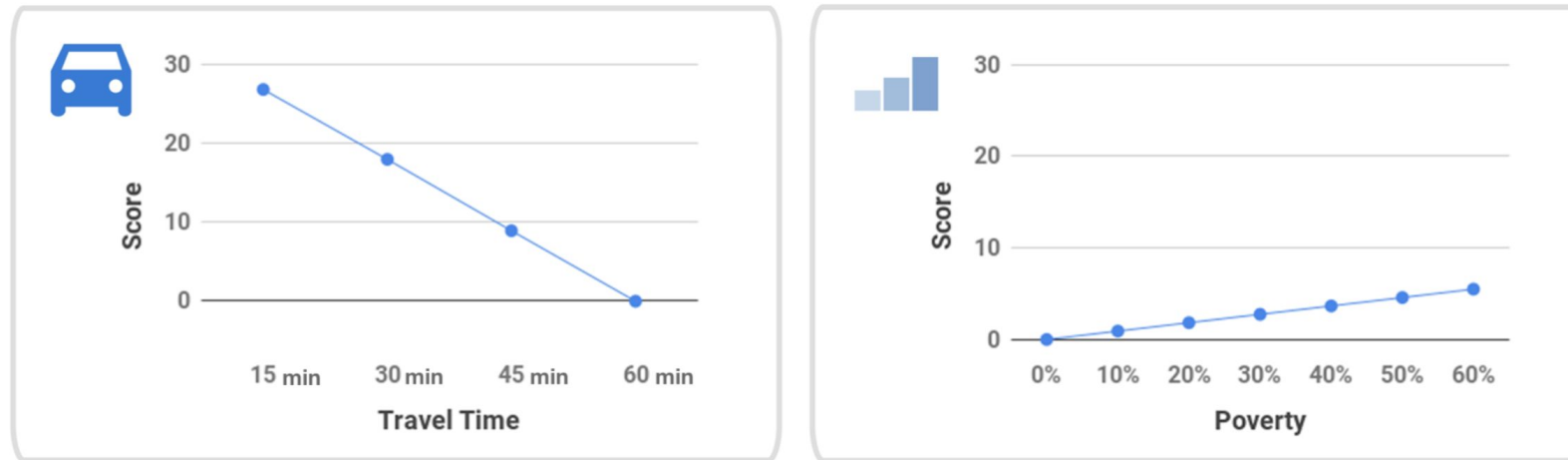
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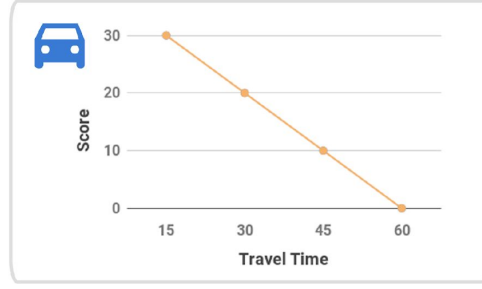
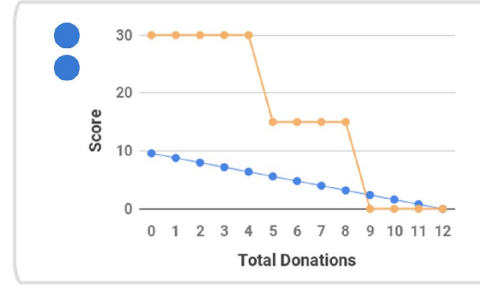
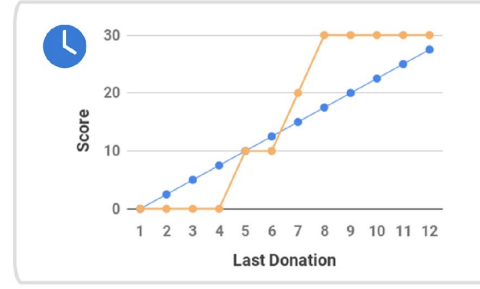
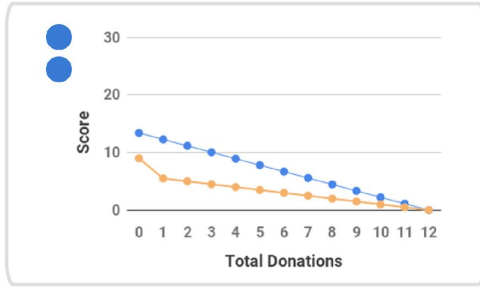
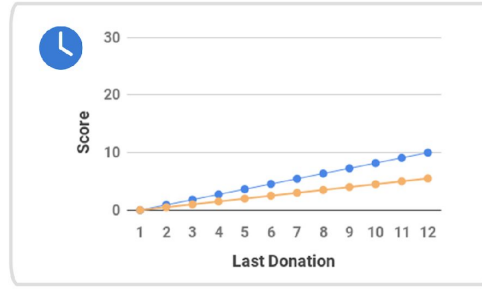
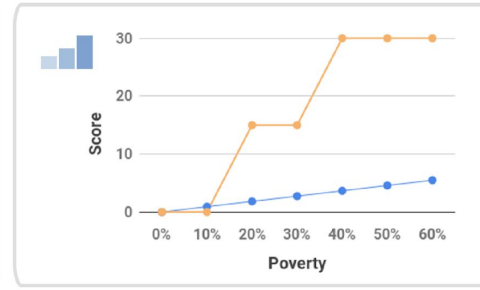
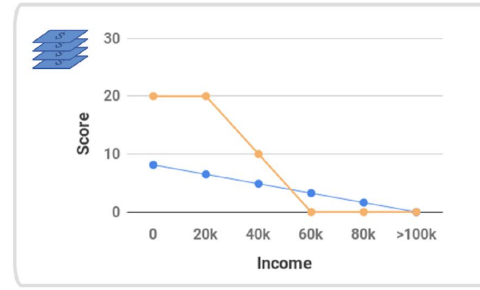
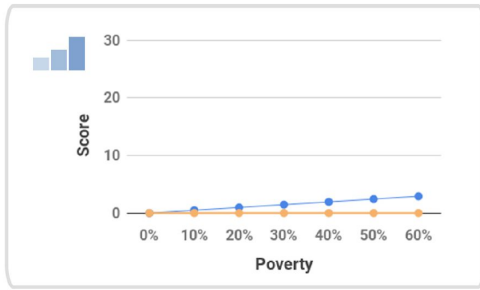
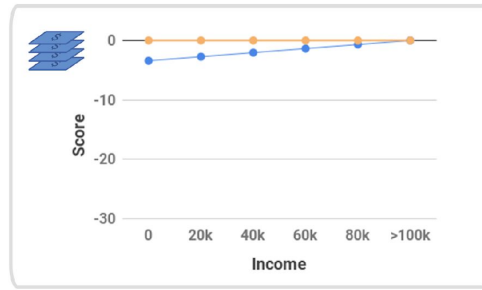
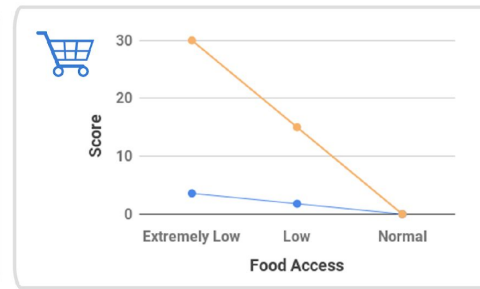
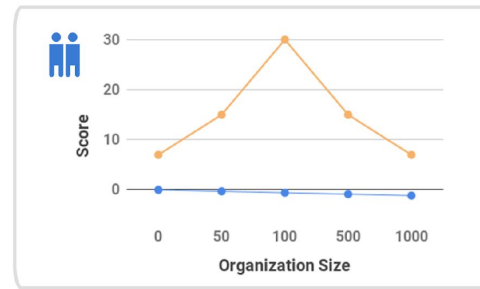
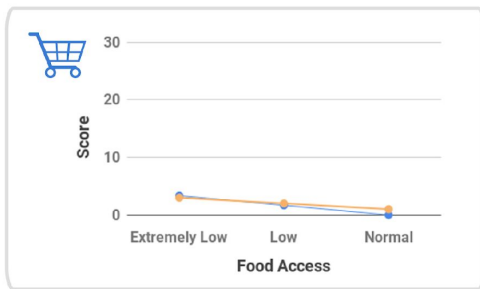
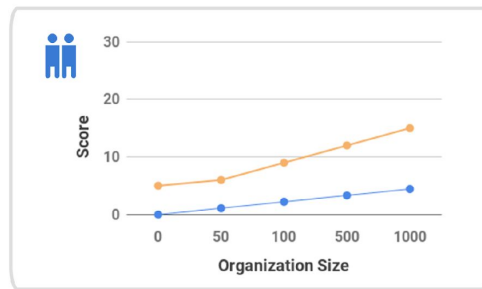
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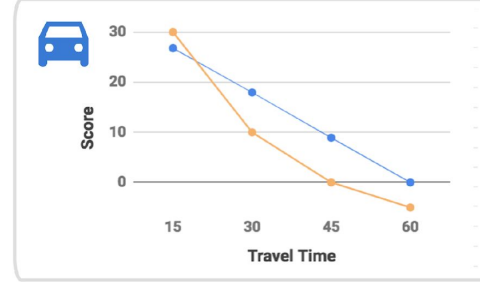
Choosing a representative model

Explaining models to choose representative model





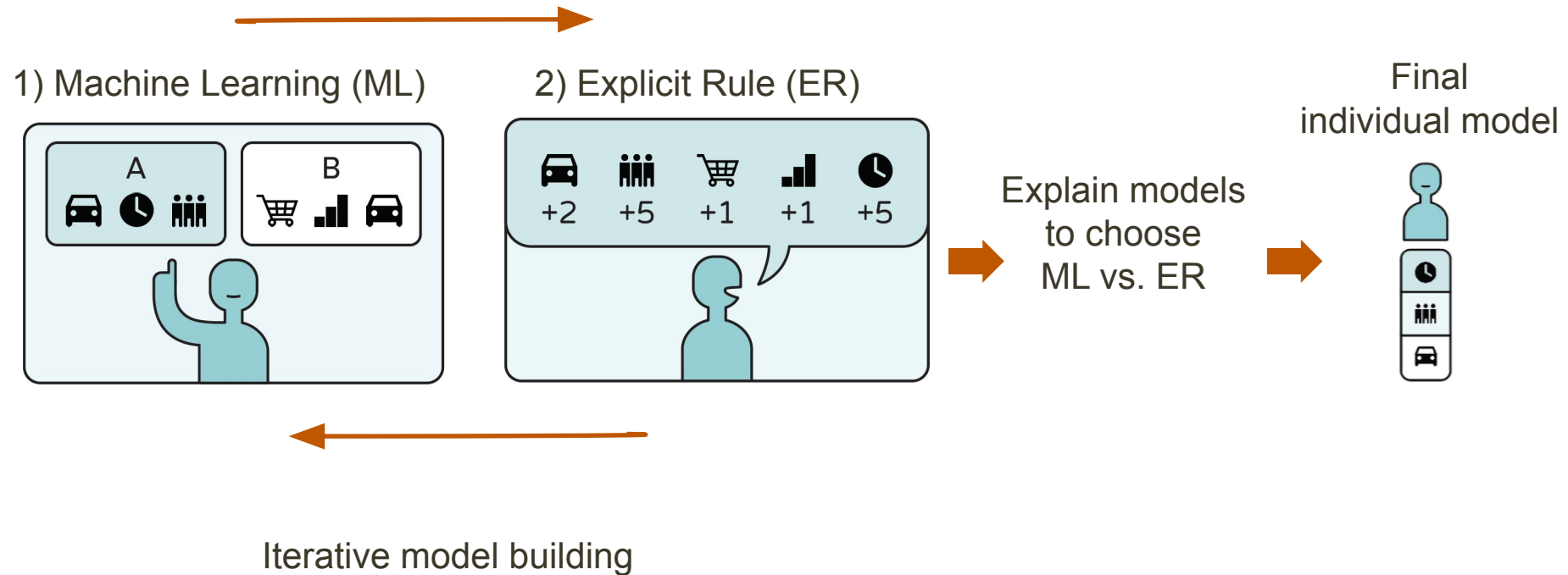
● Model X
● Model Y



● Model X
● Model Y

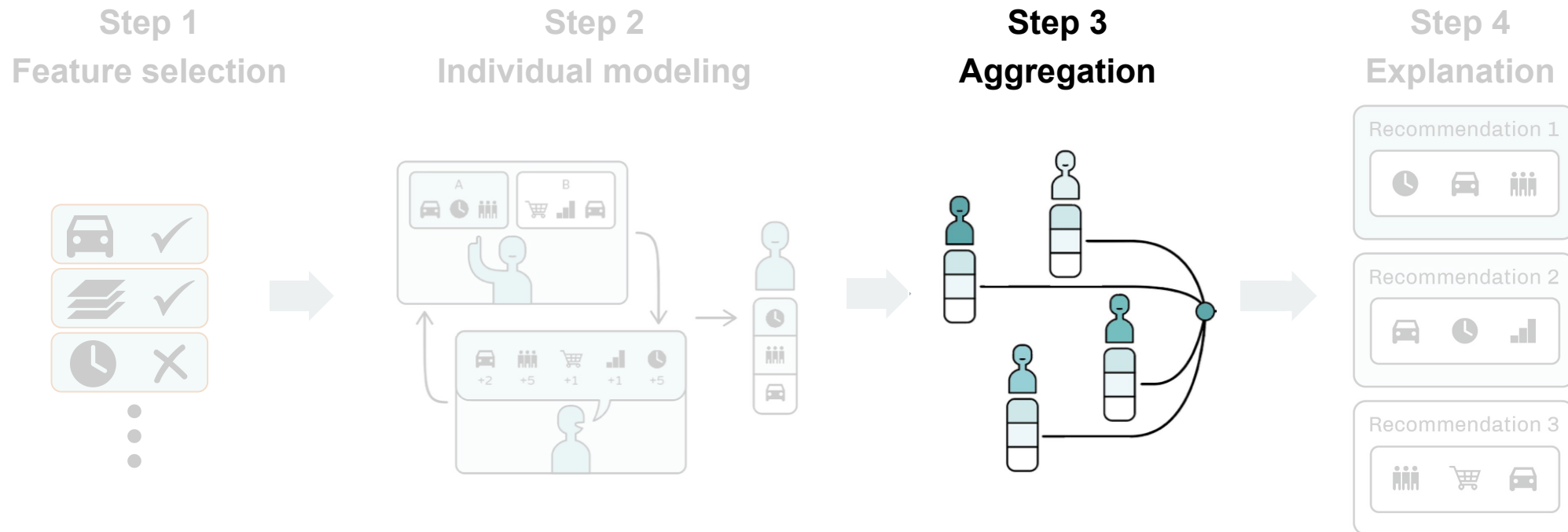
Step2: Individual model building

Used two methods iteratively to model individual's belief on how the algorithm should determine recipients



WeBuildAI: Collective participatory framework

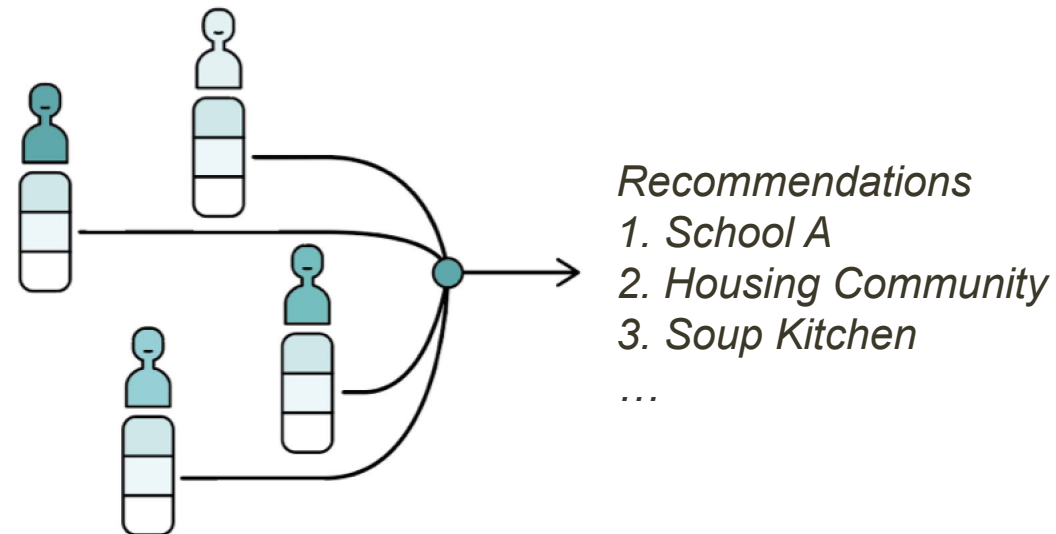
Build a computational model representing each individual stakeholder and to have that model vote on its individual's behalf



Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Step3: Collective aggregation

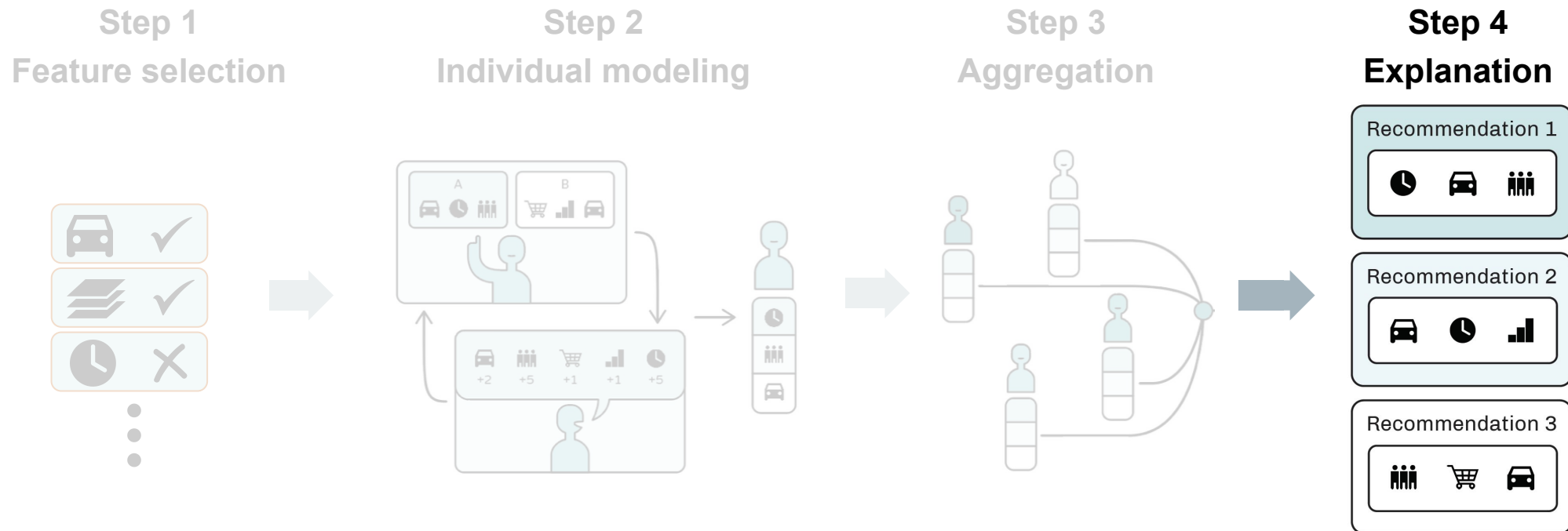
Given a new donation, each individual model generates a complete ranking of all possible recipients. The individual lists are aggregated to create recommendations



Kahng, A., Lee, M. K., Noothigattu, R., Procaccia, A. D., & Psomas, C. A. (2019) Statistical foundations of virtual democracy. ICML2019.

WeBuildAI: Collective participatory framework

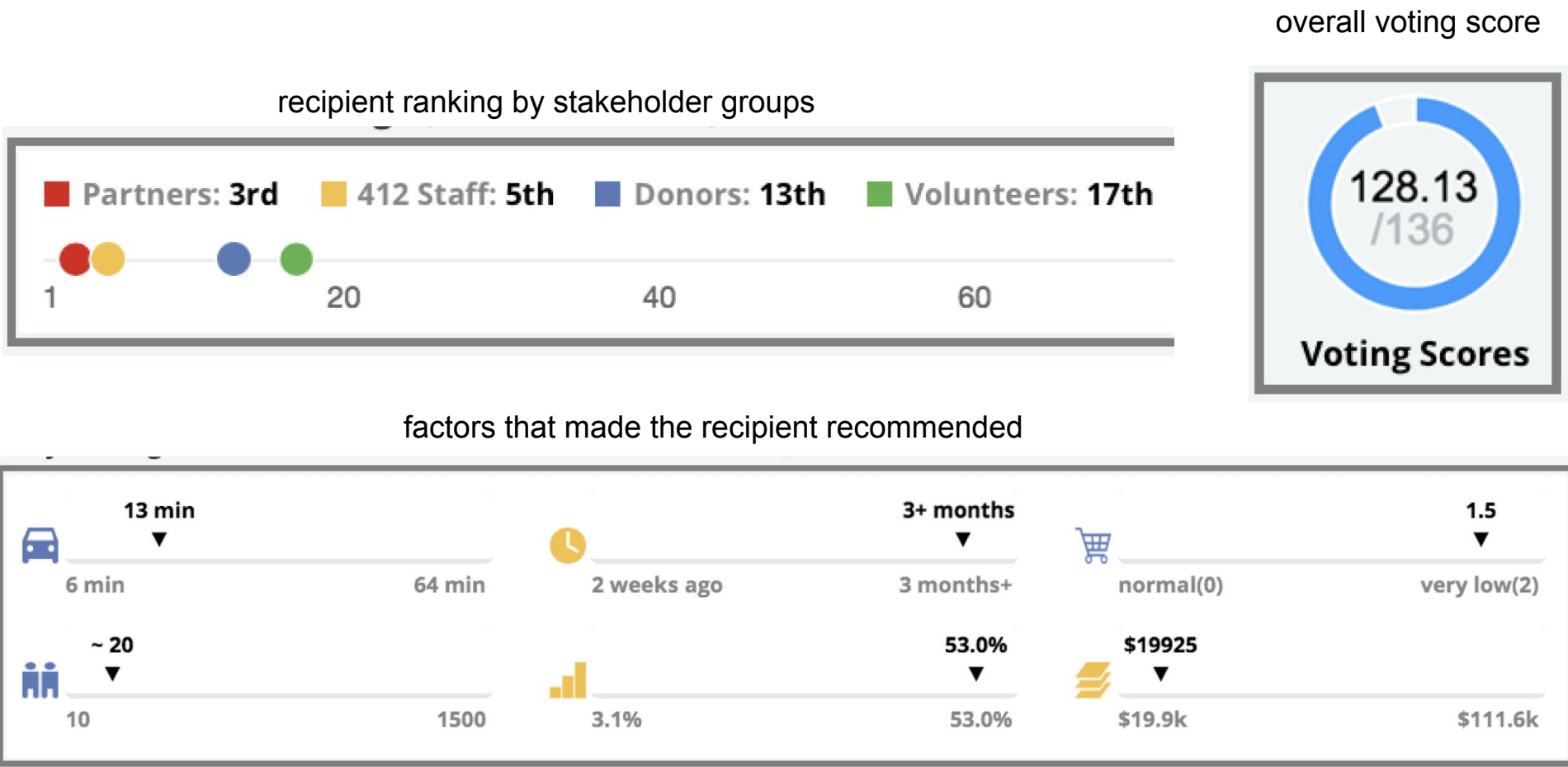
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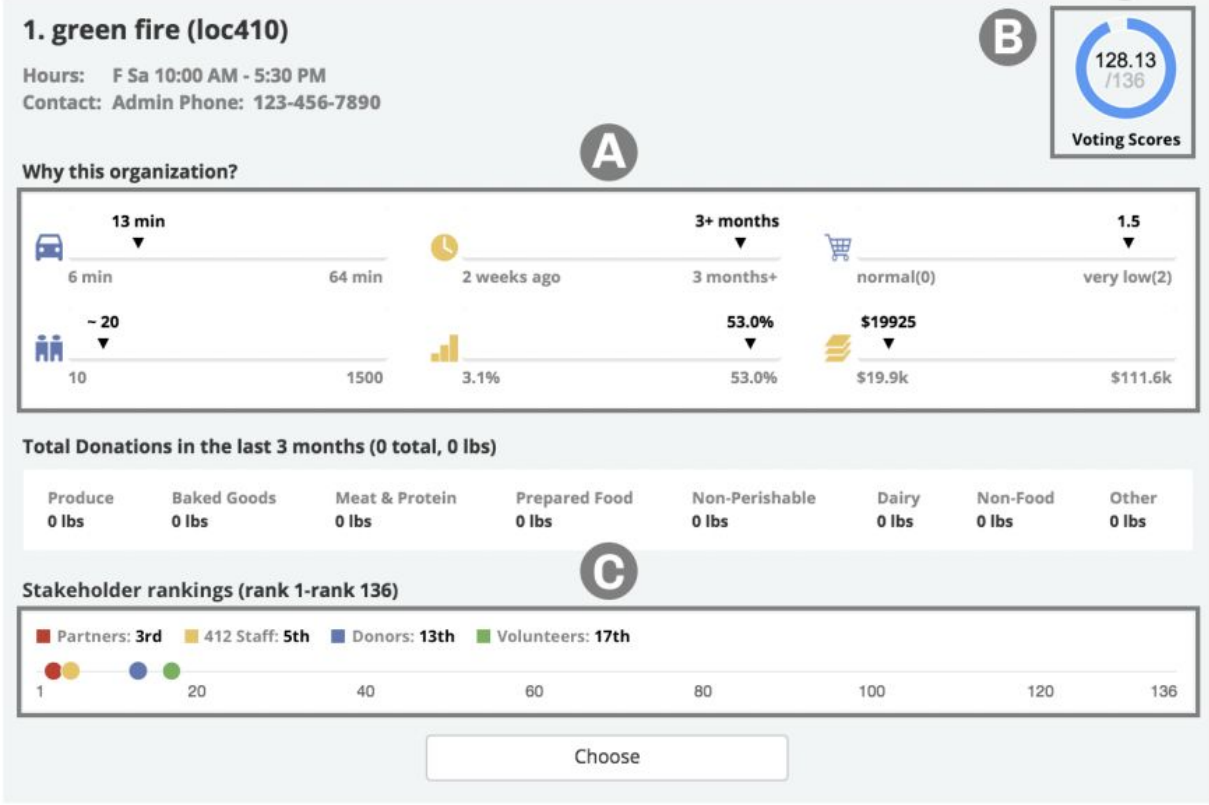


Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Step4: Explanation

Explain algorithmic recommendations to support operation





Nonprofit partner

Show recommendations

or

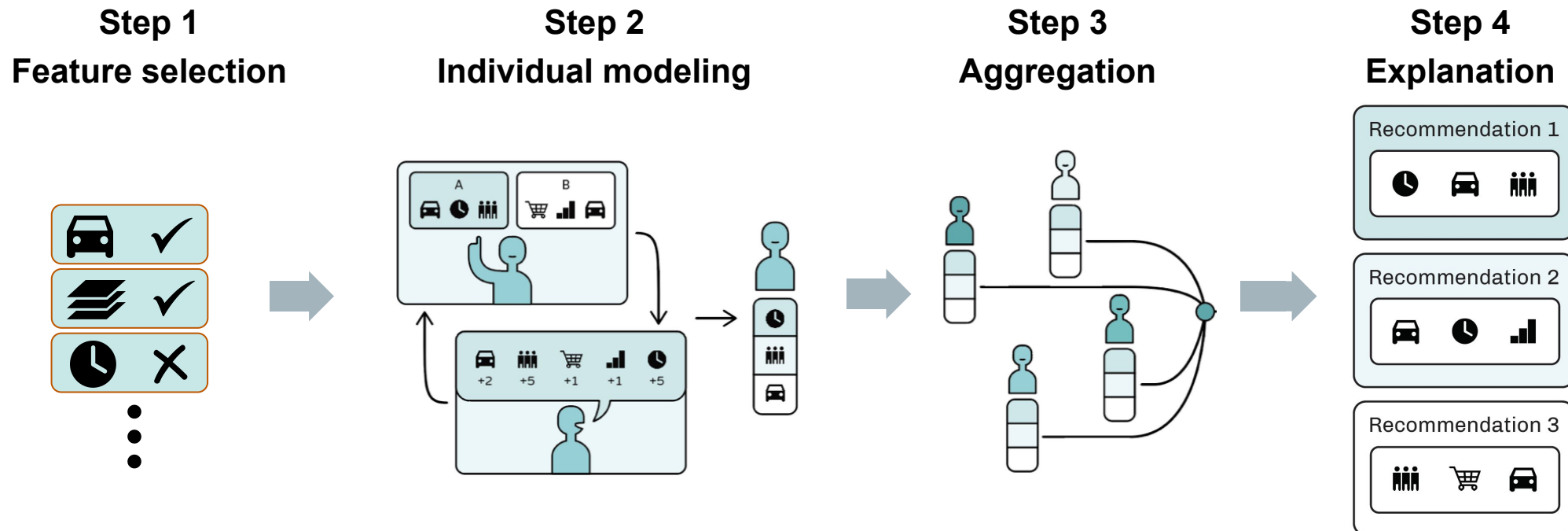
Choose



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WeBuildAI: Collective participatory framework

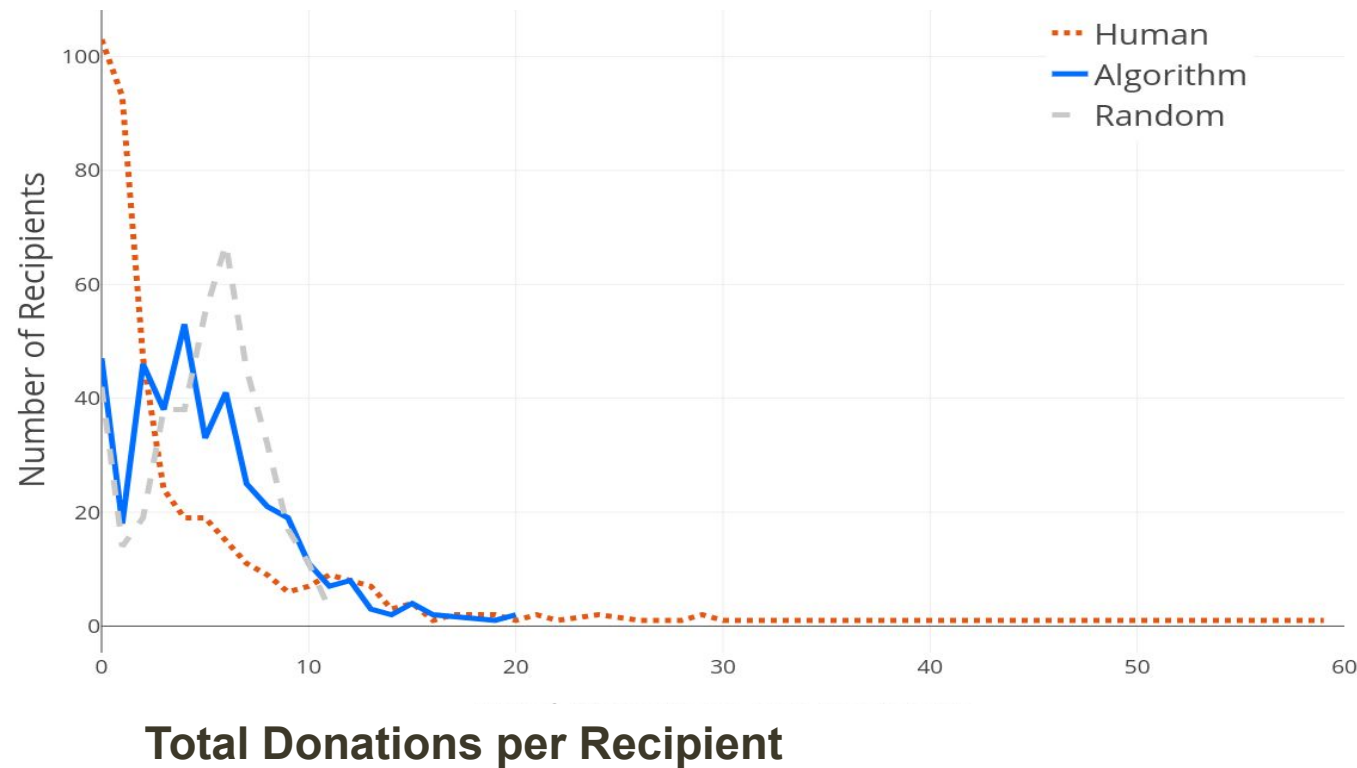
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Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

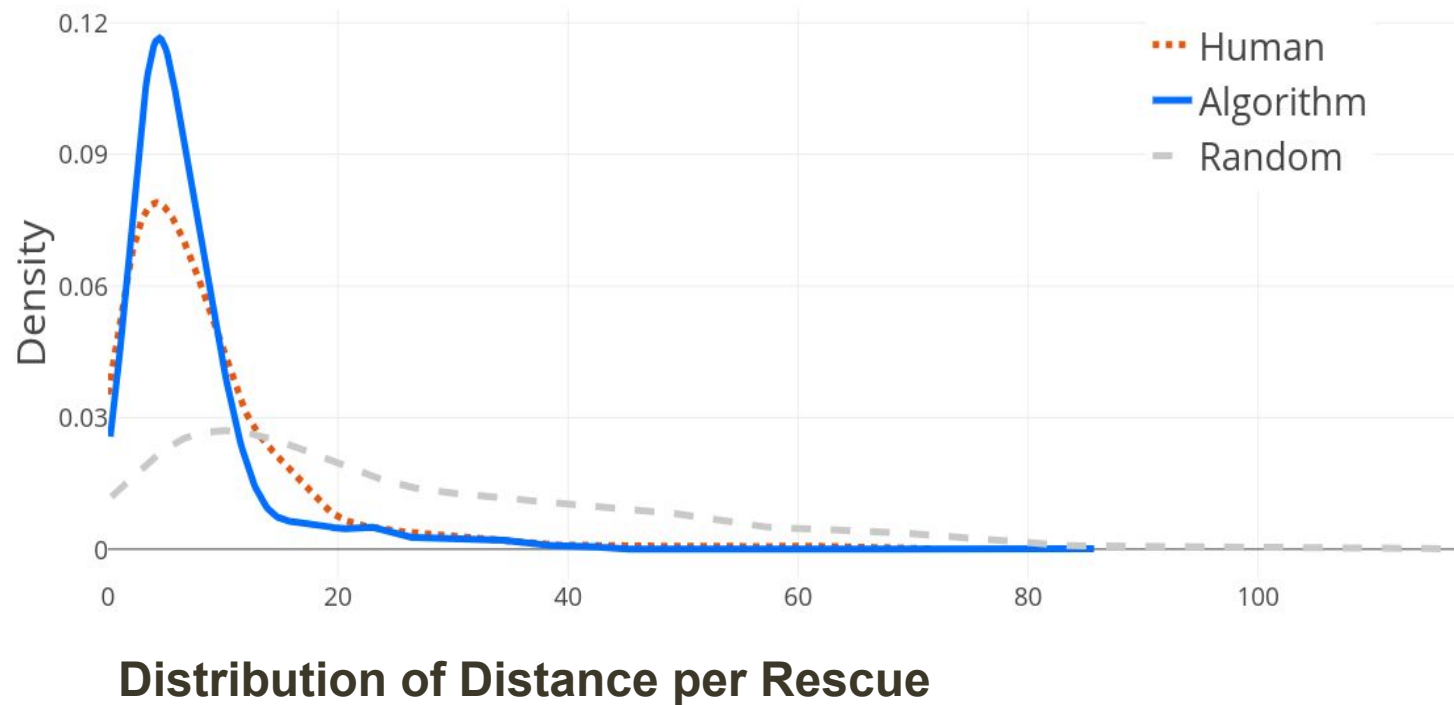
Findings: Distributive outcome

Algorithmic allocation improves equity without hurting efficiency compared to human allocation



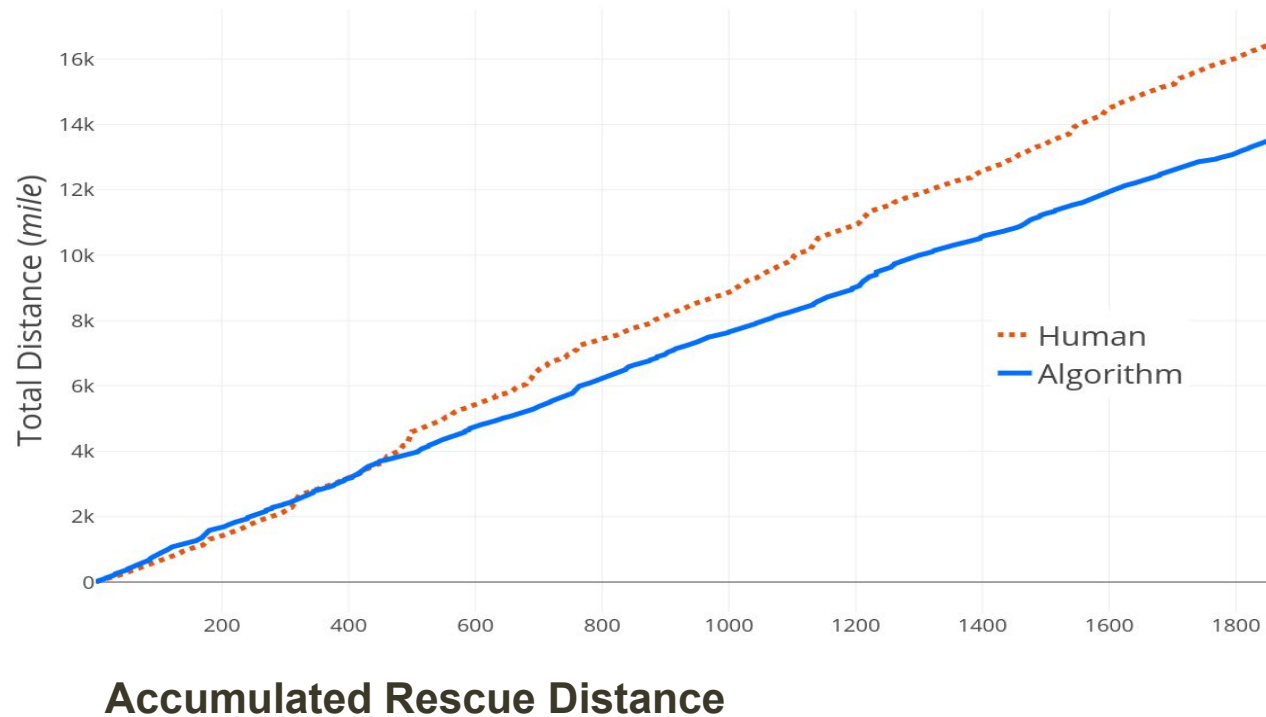
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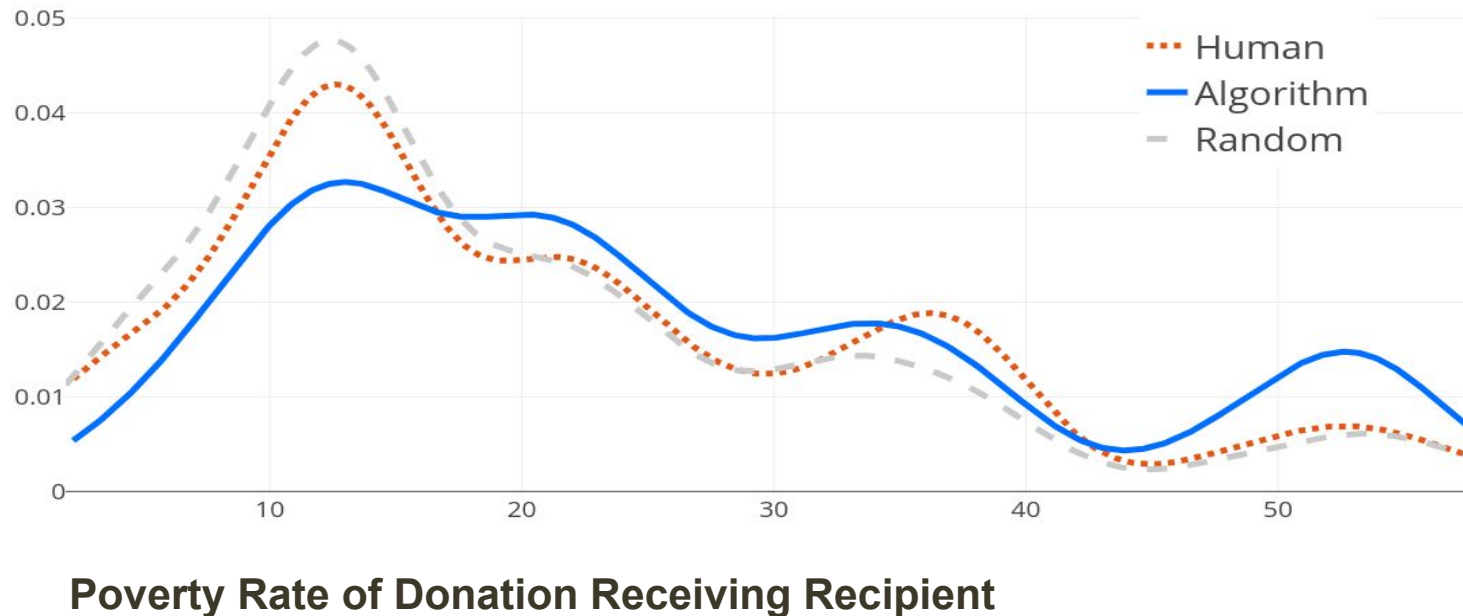
Findings: Distributive outcome

Algorithmic allocation improves equity without hurting efficiency compared to human allocation



Findings: Distributive outcome

Algorithmic allocation improves equity without hurting efficiency compared to human allocation



Findings

Legitimacy: Collective decision & algorithmic fairness

Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Collective decision & algorithmic fairness

Improved perceived fairness of the algorithm & governing organization

“These are everybody’s brain power who were deemed to be important in this decision... it should be the most fair that you could get.”

Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Findings

Legitimacy: Collective decision & algorithmic fairness

Mutual learning:

- Researchers learning how to balance efficiency and equity

- Participants' individual model & awareness

Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

Individual model & algorithmic awareness

Algorithmic capability

Difficulties in governance decision

Organizational inconsistency

Lee et al. (2019) WeBuildAI: Participatory framework for algorithmic governance. CSCW2019

In-Class Activity



The challenges of approaches like WeBuildAI

Join 2-3 peers to discuss the challenges

1. Pick a domain, and a problem that you think AI could help to resolve/mitigate
2. Discuss **two challenges** that you think you'd encounter if you were going to implement the approach of the WeBuildAI framework in the chosen domain

Submit your doc to the slack channel for this class